11. Tuned mass damper

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Modeling the tuned mass damper

▶ 13:09 / 13:09 ▸ 2.0x

Note that these two equations are both second order.

Because their highest derivative is a second derivative.

The system of equations is also inhomogeneous

because of this forcing term.

And finally, the two equations are

coupled because they both contain x1,

x1 dot, x2, and x2 dot.

In the next video, we will work on putting this system

of equations in matrix form.

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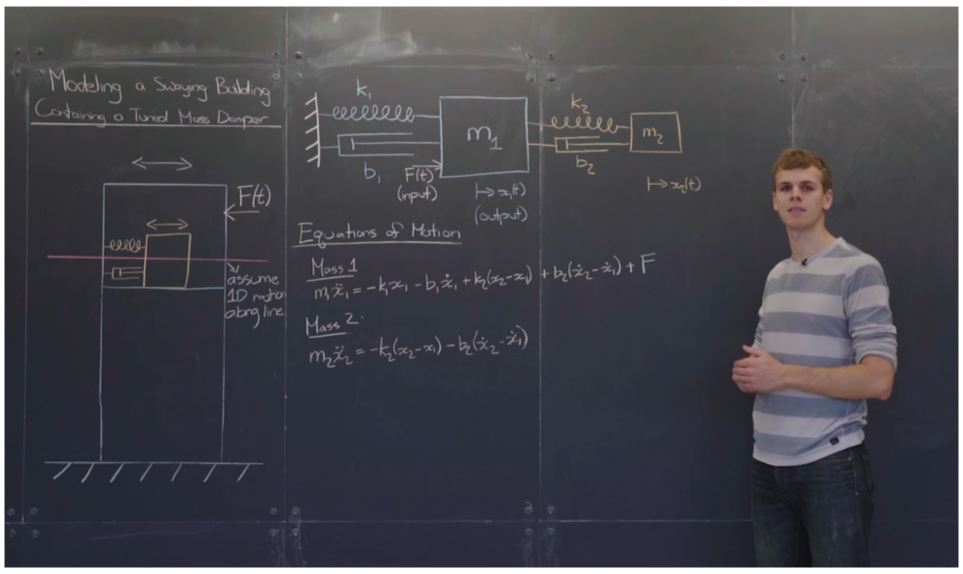
Conversion to companion system



0:06 / 11:51 ▸ 2.0x

In the previous video, we finished

modeling a couple system of oscillators consisting

of a tuned mass damper oscillating in a swaying

building.

We obtained these two equations that

are both second order differential equations

and that are coupled.

The next step is to express these equations in matrix form.

I invite you now to pause the video

and first write these two equations as a

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At the end of this lecture, you will find the normal modes for an idealized tuned mass damper that has damping and no external force acting on it. You will then solve the full inhomogeneous tuned mass damper system in the homework.

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Topic: Unit 3: Solving systems of first order ODEs using matrix methods / 11. Tuned mass damper

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